

# Mathematical and Computational Methods

## Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

This sheet accompanies *Unit 3 (Complex numbers II: exponential form, De Moivre, roots)*. It exercises the exponential form, De Moivre's theorem, the  $n$ th roots of a complex number, and the trigonometric identities they generate; nothing beyond Units 1–3 is needed.

**How to use this sheet.** Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

**Solutions.** This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit3}"`) and recompile.

**Notation.** As in the notes,  $i^2 = -1$ ,  $e^{i\theta} = \cos \theta + i \sin \theta$ , a nonzero  $z$  has exponential form  $z = r e^{i\theta}$  with  $r = |z|$  and  $\theta = \arg z$ , and  $\bar{z}$  is the complex conjugate. Give exact answers (integers, simple surds or fractions) unless a decimal is asked for.

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## 1 Warm-up drills

Short routine questions, at least one per skill. Aim for speed and accuracy; a few ask for a one- or two-line “show” derivation.

**Exercise 1.1** (★ drill— *exponential form*). Write each number in exponential form  $r e^{i\theta}$  with  $r > 0$  and  $-\pi < \theta \leq \pi$ :

$$(a) 1 + i, \quad (b) -3, \quad (c) 2i, \quad (d) -1 - i, \quad (e) \sqrt{3} - i, \quad (f) -2 + 2\sqrt{3}i.$$

**Exercise 1.2** (★ drill— *back to Cartesian*). Write each in Cartesian form  $a + ib$ :

$$(a) 2e^{i\pi/3}, \quad (b) 3e^{i\pi}, \quad (c) \sqrt{2}e^{-i\pi/4}, \quad (d) 4e^{i3\pi/2}, \quad (e) e^{i5\pi/6}.$$

**Exercise 1.3** (*★ drill— Euler values*). Evaluate  $e^{i\pi/2}$ ,  $e^{i2\pi}$ ,  $e^{-i\pi}$  and  $e^{i\pi/3}$ , and state Euler's identity.

**Exercise 1.4** (*★ drill— rotation*). Let  $z = 2e^{i\pi/6}$ . Write down  $e^{i\pi/2}z$  in exponential form, and describe geometrically the effect of multiplying a complex number by (i)  $i$ , (ii)  $e^{i\pi}$ , (iii)  $e^{-i\pi/2}$ .

**Exercise 1.5** (*★ drill— De Moivre*). Use De Moivre's theorem to simplify  $(\cos \theta + i \sin \theta)^4$  and  $(\cos \theta + i \sin \theta)^{-2}$ .

**Exercise 1.6** (*★ drill— integer powers*). Compute, using the exponential form:

$$(a) (1 + i)^4, \quad (b) (\sqrt{3} + i)^3, \quad (c) (1 - i)^6, \quad (d) (2e^{i\pi/5})^{10}.$$

**Exercise 1.7** (*★ drill— a few roots*). Find all the (a) square roots of  $i$ , and (b) fourth roots of  $1$ .

**Exercise 1.8** (*★ drill— Argand sketch*). Plot the four fourth roots of unity  $1, i, -1, -i$  on an Argand diagram, mark the unit circle, and name the regular polygon they form.

**Exercise 1.9** (*★ drill— counting roots (FTA)*). State, with the fundamental theorem of algebra in mind, how many roots in  $\mathbb{C}$  (counted with multiplicity) each has: (a) a degree-5 polynomial; (b)  $z^7 - 3 = 0$ ; (c)  $(z - 1)^2(z^2 + 1) = 0$ . True or false: every real cubic has at least one real root?

**Exercise 1.10** (*★ drill— double angle by De Moivre*). By expanding  $(\cos \theta + i \sin \theta)^2$ , derive the identities for  $\cos 2\theta$  and  $\sin 2\theta$ .

**Exercise 1.11** (*★ drill— hyperbolic link*). Using  $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ , evaluate  $\cos(i)$  in terms of  $e$ . (The general identity  $\cos(ix) = \cosh x$  is built up in the tutorial.)

**Exercise 1.12** (*★ drill— phasors*). (a) Write  $5 \cos(\omega t + \frac{\pi}{3})$  and  $2 \sin \omega t$  as phasors  $P = A e^{i\varphi}$ . (b) The phasor  $3 e^{-i\pi/2}$  represents which real signal  $A \cos(\omega t + \varphi)$ ?

## 2 Tutorial problems

The core set, to be worked through in the tutorial. Anchored on the worked examples of the notes. Problems marked ► are the core set for the 2-hour tutorial; the rest are for completion at home.

**Exercise 2.1** (► *★★ core— arithmetic in exponential form*). Let  $z = \sqrt{3} + i$  and  $w = 1 + i$ . Write  $z$  and  $w$  in exponential form, then find  $zw$  and  $z/w$  in both exponential and Cartesian form.

**Exercise 2.2** (*★★ core— rotation in action*). (a) Rotate  $z = 3 + 4i$  anticlockwise by  $\frac{\pi}{2}$  about the origin, and then (separately) clockwise by  $\frac{\pi}{2}$ . (b) Rotate  $2 + 2i$  anticlockwise by  $\frac{\pi}{4}$ . Give exact Cartesian answers.

**Exercise 2.3** (*★★ core— prove De Moivre*). Prove by induction that  $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$  for all integers  $n \geq 1$ . (Pure-maths.)

**Exercise 2.4** (► *★★ core— powers by De Moivre (anchor)*). Compute  $(1 + i)^8$  and  $(\sqrt{3} + i)^6$  using De Moivre's theorem.

**Exercise 2.5** (*★★ core— a higher power*). Evaluate  $(-1 + i)^{10}$ , and confirm your answer by squaring  $-1 + i$  first.

**Exercise 2.6** (► *★★ core— roots, two shapes (anchor)*). Find (a) the cube roots of  $8$  and (b) the fourth roots of  $-16$ . In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

**Exercise 2.7** (► \*\*core— *sixth roots of unity* (anchor)). Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all  $n \geq 2$ ?

**Exercise 2.8** (\*\*core— *solving  $w^n = z$* ). Solve  $z^3 = -27i$ , giving the three roots in exponential and Cartesian form.

**Exercise 2.9** (► \*\*core—  *$\cos 3\theta$  and  $\sin 3\theta$*  (anchor)). Using De Moivre and the binomial theorem, show that  $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$  and  $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ .

**Exercise 2.10** (\*\*core— *power reduction*). Use the inverse Euler formula to write  $\sin^4 \theta$  as a sum of cosines of multiples of  $\theta$ .

**Exercise 2.11** (► \*\*core— *trig–hyperbolic link*). (a) Show  $\cos(ix) = \cosh x$  and  $\sin(ix) = i \sinh x$ . (b) Hence evaluate  $\cos(i \ln 2)$  exactly.

**Exercise 2.12** (► \*\*core— *combining oscillations* (anchor; physics)). (a) Combine  $3 \cos \omega t + 4 \sin \omega t$  into a single sinusoid  $R \cos(\omega t - \alpha)$ . (b) Combine  $2 \cos \omega t + 2 \cos(\omega t + \frac{\pi}{2})$  likewise.

### 3 Further practice (home)

A larger bank for independent study, routine to hard, with challenge problems marked \*\*\*.

#### Euler, forms and rotation (Skills 1–2)

**Exercise 3.1** (\*drill— *mixed conversions*). (a) To exponential form ( $-\pi < \theta \leq \pi$ ):  $-4$ ,  $3i$ ,  $-2 - 2i$ ,  $-1 + \sqrt{3}i$ . (b) To Cartesian form:  $5e^{i\pi/2}$ ,  $2e^{-i2\pi/3}$ ,  $\sqrt{2}e^{i3\pi/4}$ .

**Exercise 3.2** (\*drill— *index laws*). Simplify to a single  $re^{i\theta}$  (then to Cartesian if clean): (a)  $e^{i\pi/3} \cdot e^{i\pi/6}$ , (b)  $\frac{8e^{i5\pi/6}}{2e^{i\pi/3}}$ , (c)  $(\sqrt{2}e^{i\pi/8})^4$ .

**Exercise 3.3** (\*\*core— *a rotated square*). A square is centred at the origin with one vertex at  $z_0 = 2 + i$ . Find its other three vertices by repeatedly multiplying by  $i$ , and state the side length.

**Exercise 3.4** (\*\*core— *rotation preserves geometry*). Let  $\alpha \in \mathbb{R}$ . Show that for all  $z, w \in \mathbb{C}$ ,  $|e^{i\alpha}z| = |z|$  and  $|e^{i\alpha}z - e^{i\alpha}w| = |z - w|$ . What does the second identity say geometrically?

#### De Moivre and powers (Skills 3–4)

**Exercise 3.5** (\*\*core— *more powers*). Evaluate (a)  $(1 - i\sqrt{3})^9$ , (b)  $(2 + 2i)^5$ , (c)  $\left(\frac{1+i}{\sqrt{3}+i}\right)^{12}$ .

**Exercise 3.6** (\*\*core— *De Moivre for negative  $n$* ). Assuming De Moivre for positive integers, prove it for negative integers  $n = -m$  ( $m \in \mathbb{N}$ ). [Hint:  $(\cos \theta + i \sin \theta)^{-1} = \cos \theta - i \sin \theta$ .]

#### Roots and the fundamental theorem (Skills 5–6)

**Exercise 3.7** (\*drill— *assorted roots*). Find all: (a) square roots of  $2i$ ; (b) cube roots of  $-1$ ; (c) fourth roots of  $-1$ ; (d) fifth roots of  $32$  (leave in exponential form).

**Exercise 3.8** (\*\*core— *square root the algebraic way*). The argument of  $5 - 12i$  is not a standard angle, so the exponential method is awkward. Instead find both square roots of  $5 - 12i$  in Cartesian form by equating real and imaginary parts of  $(a + ib)^2$ , and verify by squaring.

**Exercise 3.9** (\*\* core— a non-real right-hand side). Solve  $z^3 = 1 + i$ , leaving the roots in exponential form.

**Exercise 3.10** (\*\* core— a sixth-degree equation). Solve  $z^6 = 64$ , giving all six roots in Cartesian form, and describe their configuration.

**Exercise 3.11** (\*\* core— data-science flavour: roots of unity). In an  $N$ -point discrete Fourier transform the sampling frequencies are the  $N$ th roots of unity. Take  $N = 4$  and  $\omega = e^{i2\pi/4}$ . (a) List  $\omega^0, \omega^1, \omega^2, \omega^3$ . (b) Show that  $\sum_{k=0}^3 \omega^k = 0$  and  $\sum_{k=0}^3 \omega^{2k} = 0$ , whereas  $\sum_{k=0}^3 \omega^{0 \cdot k} = 4$ . [These “orthogonality” sums are what make the DFT invertible.]

**Exercise 3.12** (\*\* core— data-science flavour: twiddle factors). The fast Fourier transform is built from the “twiddle factors”  $W_N^k = e^{-i2\pi k/N}$ . Take  $N = 8$ . (a) Write  $W_8^1$  in Cartesian form. (b) Show that  $W_8^{k+4} = -W_8^k$  for every integer  $k$  (the symmetry that halves the FFT’s work). (c) Give the modulus and argument of  $W_8^3$ .

**Exercise 3.13** (\*\* core— FTA and real factors). (a) Verify that  $z = 2i$  is a root of  $z^2 + 4$ , and write down the other. (b) Factor  $z^4 + 4$  into two real quadratics. [Hint: find the four roots first.]

**Exercise 3.14** (\*\*\* challenge— a root-of-unity product). Let  $\omega = e^{i2\pi/5}$ . Show that  $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) = 5$ . Generalise to  $n$ .

**Exercise 3.15** (\*\*\* challenge— a conjugate equation). Find all  $z \in \mathbb{C}$  with  $z^2 = \bar{z}$ .

### Trigonometric identities (Skill 7)

**Exercise 3.16** (\*\* core—  $\cos 4\theta$ ). Use De Moivre to show  $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$ .

**Exercise 3.17** (\*\* core—  $\cos^5 \theta$ ). Use the inverse Euler formula to express  $\cos^5 \theta$  as a sum of cosines of multiples of  $\theta$ .

**Exercise 3.18** (\*\*\* challenge—  $\tan 3\theta$ ). Show that  $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ .

**Exercise 3.19** (\*\*\* challenge— a finite cosine sum). Using the geometric series for  $\sum_{k=0}^{n-1} e^{ik\theta}$  (with  $\theta \neq 0 \pmod{2\pi}$ ), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{(n-1)\theta}{2}\right).$$

### Trigonometric–hyperbolic link (Skill 8)

**Exercise 3.20** (\*\* core— the converse pair). Show that  $\cosh(ix) = \cos x$  and  $\sinh(ix) = i \sin x$ , and evaluate  $\sin(2i)$  and  $\cos(\pi i)$  in terms of  $e$ .

**Exercise 3.21** (\*\* core— identity transfer). Starting from  $\cos 2\theta = 1 - 2 \sin^2 \theta$ , set  $\theta = ix$  and use the link of the previous exercise to derive the hyperbolic identity  $\cosh 2x = 1 + 2 \sinh^2 x$ .

### Phasors (Skill 9)

**Exercise 3.22** (\*\* core— a 5–12–13 combination). Combine  $5 \cos \omega t + 12 \sin \omega t$  into  $R \cos(\omega t - \alpha)$ , giving  $R$  exactly and  $\alpha$  to three decimals.

**Exercise 3.23** (\*\* core— physics word problem: two AC sources). Two alternating voltages of equal amplitude but different phase,  $V_1 = 10 \cos \omega t$  and  $V_2 = 10 \cos(\omega t + \frac{2\pi}{3})$  (volts), are added. Find the amplitude and phase of  $V_1 + V_2$ .

**Exercise 3.24** ( $\star\star\star$  challenge— *balanced three-phase*). Show that three equal phasors  $120^\circ$  apart sum to zero:  $1 + e^{i2\pi/3} + e^{i4\pi/3} = 0$ . Interpret the result physically.

**Concept checks (true/false and spot-the-error)**

**Exercise 3.25** ( $\star$  drill— *true or false*). Decide and justify briefly: (a)  $|e^{i\theta}| = 1$  for every real  $\theta$ . (b)  $e^{i\theta} = e^{i\varphi}$  implies  $\theta = \varphi$ . (c) Every complex number has exactly three cube roots. (d) The  $n$ th roots of unity always include 1.

**Exercise 3.26** ( $\star\star$  core— *spot the error*). A student writes: “Since  $(\cos \theta + i \sin \theta)^{1/2} = \cos \frac{\theta}{2} + i \sin \frac{\theta}{2}$  by De Moivre, the square root of  $e^{i\theta}$  is  $e^{i\theta/2}$ .” What is wrong, and what is the correct statement?